WEEK 3 · LESSON 17 OF 24 · HANDOUT

# Distributions and the Normal Curve

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=bPFNxD3Yg6U — open the lesson online to watch inside the course.

**Your goal in this lesson:** Recognise common shapes of data (normal, skewed, uniform, bimodal), use the 68–95–99.7 rule, compute z-scores, and standardise features for AI.

## The shape of data

A **distribution** describes how often each value occurs. Draw a histogram and you see its shape:

| Shape | What it looks like | Example |
|---|---|---|
| **Normal** (bell curve) | Symmetric, most values near the middle | Adult heights, measurement errors |
| **Right-skewed** | A long tail to the right | Incomes, house prices, sale sizes |
| **Left-skewed** | A long tail to the left | Marks on a very easy test |
| **Uniform** | All values equally common | A fair die |
| **Bimodal** | Two peaks | Arrival times at a bus stop with two rush hours |

In a right-skewed distribution, the **mean is larger than the median** — the long tail pulls the mean up.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(1)
normal = rng.normal(165, 8, 5000)            # heights in cm
skewed = rng.lognormal(12, 0.6, 5000)        # incomes, long right tail
uniform = rng.integers(1, 7, 5000)           # dice

fig, axes = plt.subplots(1, 3, figsize=(12, 3))
axes[0].hist(normal, bins=40); axes[0].set_title("Normal")
axes[1].hist(skewed, bins=40); axes[1].set_title("Right-skewed")
axes[2].hist(uniform, bins=np.arange(0.5, 7.5, 1)); axes[2].set_title("Uniform")
plt.tight_layout()
plt.show()
print(np.mean(skewed) > np.median(skewed))   # True
```

## The normal distribution

The normal (Gaussian) distribution is described by just two numbers: the **mean (μ)** and the **standard deviation (σ)**. It appears everywhere because many small random effects added together produce a bell curve.

**The 68–95–99.7 rule:** in a normal distribution, - about **68%** of values are within 1 SD of the mean, - about **95%** within 2 SD, - about **99.7%** within 3 SD.

If adult heights have mean 165 cm and SD 8 cm, about 95% of adults are between 149 and 181 cm.

```
heights = rng.normal(165, 8, 100000)
within_2sd = ((heights > 165 - 16) & (heights < 165 + 16)).mean()
print(round(within_2sd, 3))   # about 0.954
```

## z-scores: how unusual is a value?

A **z-score** says how many standard deviations a value is from the mean:

> *z = (value − mean) ÷ SD*

A student scores 82 in a test with mean 70 and SD 6: z = (82 − 70) ÷ 6 = **2.0** — better than about 97.7% of the class (if marks are roughly normal). A z-score above 3 or below −3 is very unusual and worth checking.

z-scores let you compare different scales: a 75 in a hard exam may be more impressive than an 85 in an easy one.

## Standardisation for AI

Features often have very different scales: size in m² (50–300), bedrooms (1–6), price in RWF (millions). Many models (k-NN, neural networks, gradient descent) work much better when every feature is on a similar scale. **Standardising** turns each feature into z-scores — mean 0, SD 1:

```
X = np.array([[120, 3], [80, 2], [200, 4], [95, 2]], dtype=float)
X_std = (X - X.mean(axis=0)) / X.std(axis=0)
print(X_std.round(2))
print(X_std.mean(axis=0).round(2), X_std.std(axis=0).round(2))   # ~[0 0] [1 1]
```

You will use this in Week 4 to make gradient descent work well.

## Key terms

- **Distribution** — how often each value occurs.
- **Normal distribution** — the symmetric bell curve, described by μ and σ.
- **Skewed** — having a long tail on one side.
- **68–95–99.7 rule** — shares of a normal distribution within 1, 2 and 3 SDs.
- **z-score** — (value − mean) ÷ SD.
- **Standardisation** — rescaling features to mean 0 and SD 1.

**Remember this:** Always look at the shape of your data. Use the 68–95–99.7 rule for bell-shaped data, z-scores to measure how unusual a value is, and standardisation to put features on the same scale.

Name: _____ Date: _____

## Activity

### Task 1

Simulate 10,000 test marks with mean 60 and SD 12. What share are above 84? Compare with the 68–95–99.7 rule.

### Task 2

A sale of 45,000 RWF happens in a shop where sales have mean 12,000 and SD 8,000. Compute the z-score. Is it unusual?

### Task 3

Make a histogram of the right-skewed incomes and mark the mean and median with `plt.axvline`. Which is bigger?

## Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Incomes usually have a long tail to the right. In such data, how do mean and median compare?
   A. Median is always zero
   B. They are always equal
   C. Mean > median
   D. Mean < median

2. In a normal distribution, about what share of values lies within 2 SD of the mean?

A. 99.7%
B. 95%
C. 68%
D. 50%

3. Mean 70, SD 6. What is the z-score of 82?

A. 1.17
B. -2
C. 2
D. 12

4. What does standardising a feature do?

A. Deletes outliers
B. Turns it into text
C. Sorts it
D. Rescales it to mean 0 and SD 1

5. Which two numbers fully describe a normal distribution?

A. Count and sum
B. Mean and standard deviation
C. Median and mode
D. Min and max

6. Why standardise features before training many models?

A. To make the data secret
B. Models cannot read decimals
C. It is never useful
D. So features with large numbers do not dominate and training works better

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